BJT small-signal I

Practice questions · Day 10 · Mon 2 Nov · pack E6

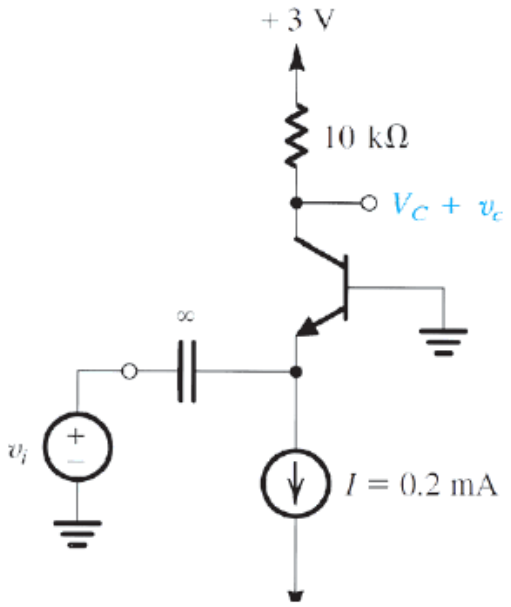
5. Practice

From your course files

WS6 Q1 (course question). A BJT having $\beta = 100$ is biased at a DC collector current of 1 mA. Find the value of g_m, r_e, and r_π at the bias point.

WS6 Q2 (course question). Given $g_m = 20$ mA/V and $\beta = 100$, find r_π and r_e. Then, draw four small-signal equivalent circuit models.

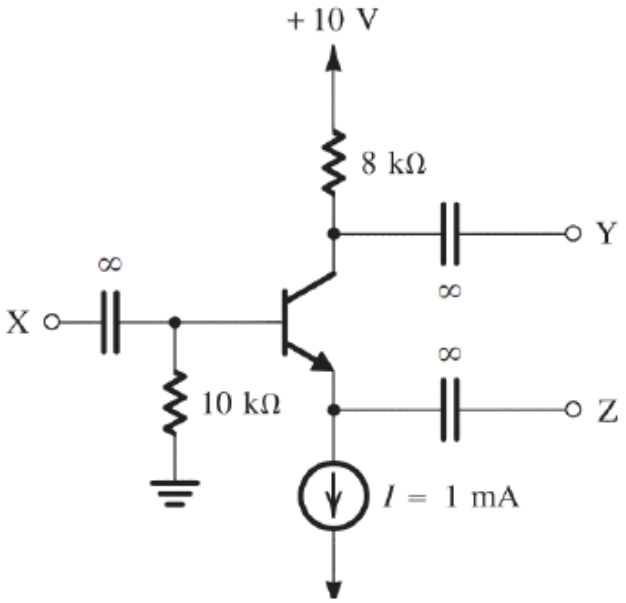
WS6 Q4 (course question). The transistor amplifier shown in the figure 9 [the figure below; the sheet labels it Figure 4] is biased with a current source $I = 0.2$ mA and has a very high β. Assuming the transistor is in the active state, the DC voltage of the collector is 1 V, and $r_e = 125\ \Omega$. Replace the transistor with the **T model** and find voltage gain, $\frac{v_c}{v_i}$.



WS6 Q4 figure (Figure 4 of the workshop sheet)

WS6 Q5 (course question). The transistor amplifier shown in the figure 10 [the figure below; the sheet labels it Figure 5] is biased with a current source $I = 1$ mA and has $\beta = 100$ and $V_A = 100$ V. Assuming that $g_m = 39.604$ mA/V, $r_\pi = 2.525\ \text{k}\Omega$ and $r_o = 101\ \text{k}\Omega$, answer the following questions:

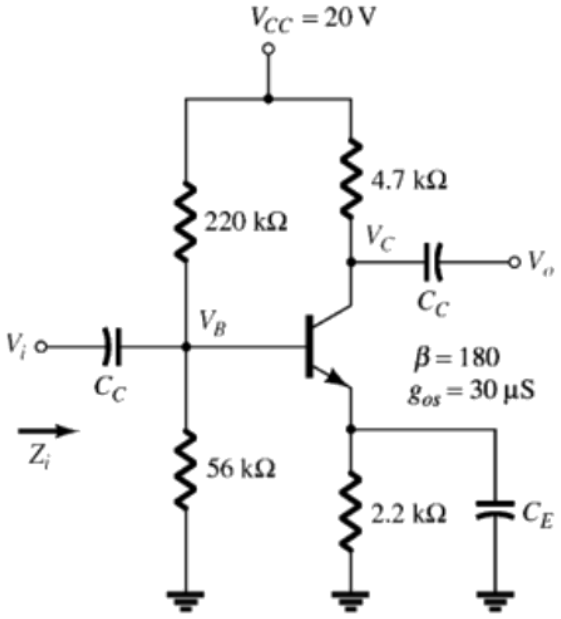
1. If terminal Z is connected to ground, X to a signal source v_{sig} with a source resistance $R_{sig} = 2\ \text{k}\Omega$, and Y to an 8-kΩ load resistance, use the hybrid-π model to draw the small-signal equivalent circuit of the amplifier.
2. Calculate the overall voltage gain, $\dfrac{v_y}{v_{sig}}$.



WS6 Q5 figure (Figure 5 of the workshop sheet)

WS6 Q6 (course question). For the network of figure 6:

1. Determine r_e.
2. Calculate V_B and V_C.
3. Calculate Z_i and $A_V = V_o/V_i$.

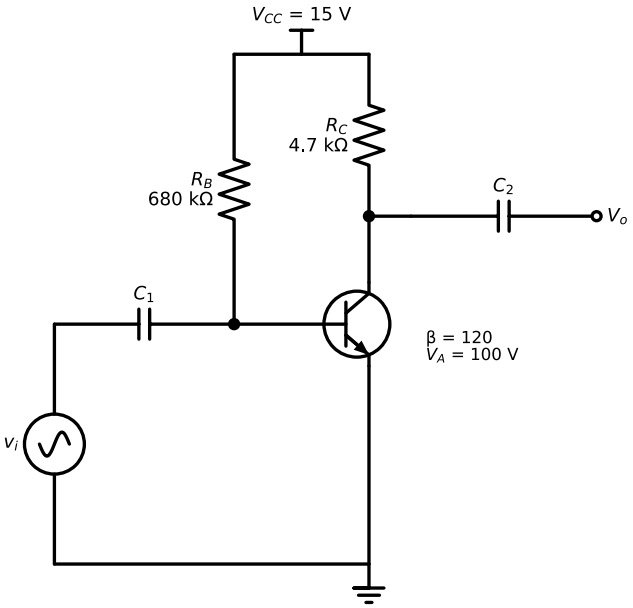


WS6 Q6 figure (Figure 6 of the workshop sheet)

New questions

N1 (easy) [4 marks]. A BJT with $\beta = 150$ and $V_A = 80$ V is biased at $I_C = 2$ mA. Take $V_T = 26$ mV. (a) Find g_m, r_π, r_e and r_o. [3] (b) Show numerically that $r_\pi = (\beta + 1) r_e$. [1]

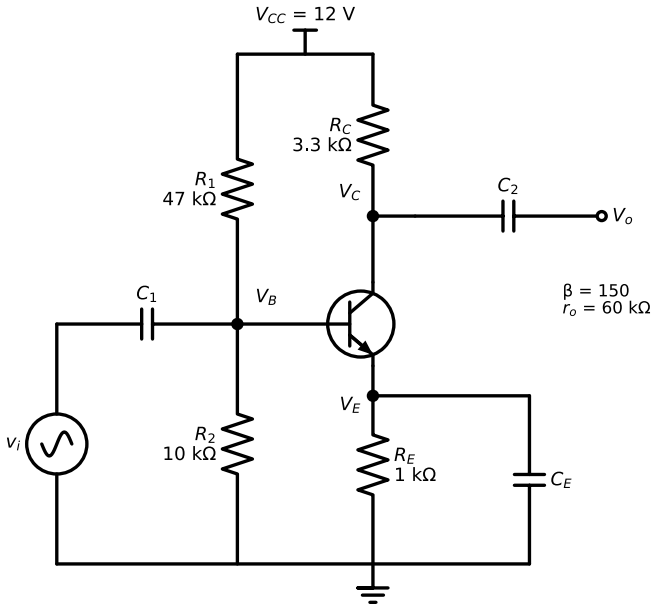
N2 (medium) [7 marks]. CE fixed-bias amplifier: $V_{CC} = 15$ V, $R_B = 680$ kΩ, $R_C = 4.7$ kΩ, $\beta = 120$, $V_A = 100$ V, ideal coupling capacitors. (a) Find the Q-point (I_B, I_C, V_{CE}) and confirm active mode. [2] (b) Find r_e and r_o. [1] (c) Find Z_i, Z_o and A_v with and without r_o. [3] (d) $v_i = 5 \sin \omega t$ mV. Write $v_o(t)$ and say whether the output will clip. [1]



N2: CE fixed-bias amplifier

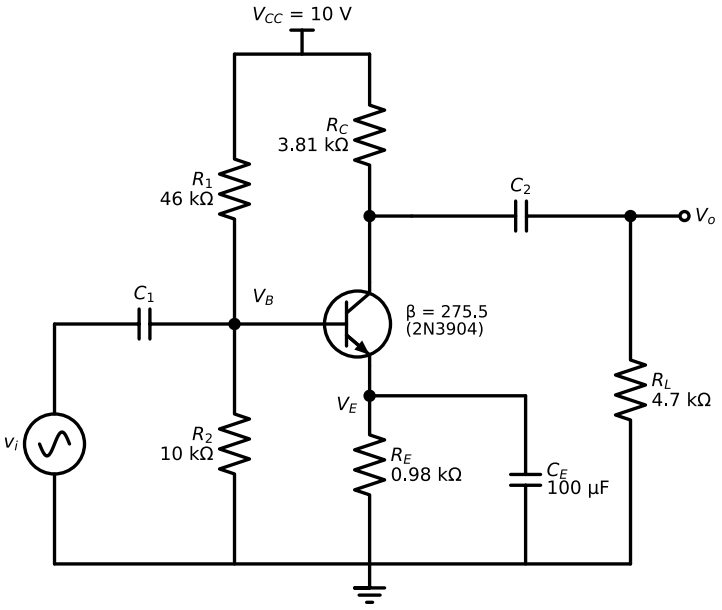
N3 (medium) [8 marks]. Voltage-divider CE with bypassed emitter: $V_{CC} = 12$ V, $R_1 = 47$ kΩ, $R_2 = 10$ kΩ, $R_C = 3.3$ kΩ, $R_E = 1$ kΩ with C_E across it, $\beta = 150$, $r_o = 60$ kΩ. (a) Test

whether the approximate method may be used, and find I_E, V_C and V_{CE}. [3] (b) Find r_e. [1] (c) Find Z_i, Z_o and A_v (include r_o). [3] (d) Write the approximate h-parameters h_{ie}, h_{fe}, h_{oe} of the transistor at this Q-point. [1]



N3: voltage-divider CE with bypassed emitter

N4 (multi-part, chains) [9 marks] — your Practical 3 amplifier. As built: $V_{CC} = 10$ V, $R_1 = 46$ kΩ, $R_2 = 10$ kΩ, $R_C = 3.81$ kΩ, $R_E = 0.98$ kΩ bypassed by 100 µF, 2N3904 with measured $\beta = 275.5$, $V_A \approx 100$ V. (a) Predict V_B, I_E with the approximate method, and compare with your measured $V_B = 1.735$ V and $I_E = 1.12$ mA. [2] (b) Using the measured $I_E = 1.12$ mA, find r_e and r_o. [1] (c) Find Z_i, Z_o and the no-load A_v, first with $r_o = \infty$ and then with r_o. [3] (d) You measured $A_v = 142.2$ (magnitude). Give one reason, other than r_o, why the measurement is lower. [1] (e) A 4.7 kΩ load is connected via C_2. It appears in parallel with $R_C \parallel r_o$. Predict the loaded gain and compare with the measured 77.1. [2]



N4: your Practical 3 amplifier as built, with the 4.7 kΩ load on C_2

N5 (hard, design) [8 marks]. (a) For any CE stage with bypassed emitter and $r_o = \infty$, show that $|A_v| = V_{R_C}/V_T$, where V_{R_C} is the DC voltage across R_C. [2] (b) A fixed-bias CE stage runs from $V_{CC} = 12$ V with $\beta = 100$. What is the largest $|A_v|$ possible if the design keeps $V_{CEQ} = 6$ V? [1] (c) Design the stage ($I_C = 1$ mA) for $A_v = -150$ (use $|A_v| = g_m R_C$): find R_C, V_{CEQ} and R_B. [3] (d) Recalculate A_v using r_e (from I_E) and with $V_A = 100$ V, and comment. [2]

6. Quick recall check

1. Write g_m, r_π, r_e and r_o in terms of the Q-point quantities.
2. In the AC equivalent of a VDB amplifier, where do R_1, R_C and the bypassed R_E go?
3. Why is $g_m v_\pi$ the same as βi_b?
4. Bypassed VDB stage: give Z_i, Z_o and A_v (with r_o).
5. A CE stage has $V_{CC} = 10$ V and $V_C = 5$ V at Q. Estimate $|A_v|$ in one line.